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Americas Clean Technology: Energy Storage: Assessing the BTM battery storage opportunity for data centers; updating ESS S/D model

Brian Lee, CFA
+1(917)343-3110 | brian.k.lee@gs.com
Goldman Sachs & Co. LLC

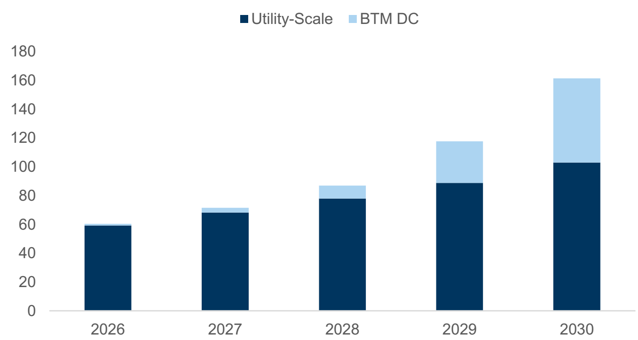
Tyler Bisset, CFA
+1(212)357-5510 | tyler.bisset@gs.com
Goldman Sachs & Co. LLC

Keshav Choudhary
+1(332)245-7971 |
keshav.choudhary@gs.com
Goldman Sachs India SPL

Battery storage is increasingly being viewed as an ideal solution to provide power to data centers, particular those focused on artificial intelligence (AI) applications, in behind-the-meter (BTM) settings as a result of several competitive advantages such as speed to market and rapid response times. Against this backdrop of a pending inflection for data center deployments, we assess the market potential for battery energy storage systems (BESS) for BTM applications and highlight companies across the battery storage supply chain to gain exposure to this theme. We also update forecasts within our battery storage supply and demand model to more directly reflect the BTM BESS opportunity in the US, as well as some other adjustments in areas like Australia and EMEA, which are experiencing strong demand growth. Lastly, we discuss BTM storage implications to both FLNC and NRGV. We provide more detailed takeaways here within.

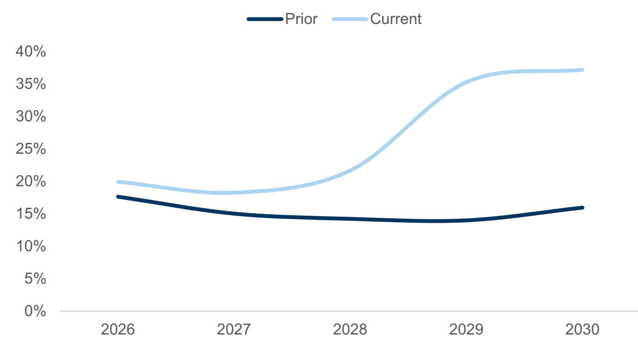
The behind the meter (BTM) battery opportunity in focus. Our Asia battery team recently outlined estimates for BTM opportunities within the US. In particular, the team estimates a ~50 GWh uplift from BTM opportunities in 2030, in addition to 11 GWh from 800V DC data center market. This translates to a utility-scale BESS deployment opportunity in the US of 161 GWh in 2030, with total US BESS deployments of 172 GWh. Importantly, our Utilities team also recently raised its US power demand CAGR to 3.2% from 2.6%, mostly driven by increased data center power demand estimates. This includes increasing on the grid power demand CAGR to 2.8%, while BTM power represents the bridge to the 3.2% CAGR (Exhibit 3), which represents 20 GW of new power capacity (14 GW assuming ~70% capacity factor) by 2030 and mostly via onsite gas. We note that the power demand inflection for data centers is happening now, especially as hyperscalers aggressively shift to AI-optimized hardware, providing an immediate opportunity for BESS.

Exhibit 1: BTM opportunity for battery storage offers strong growth potential
GS forecast for US utility-scale and BTM battery storage deployments (GWh)



Source: Goldman Sachs Global Investment Research

Exhibit 2: BTM storage has meaningfully improved the growth rate for US utility-scale storage
Year-over-year growth rates for utility-scale storage (GS forecast)

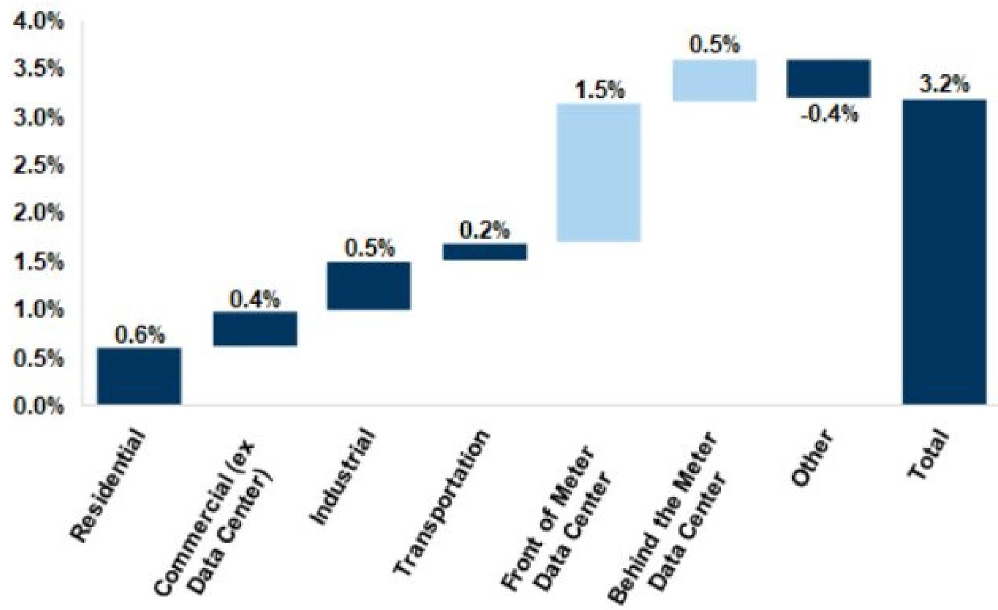


Source: Goldman Sachs Global Investment Research

Why do batteries matter for the power grid… A battery energy storage system (BESS) is a technological device that can harness electrical energy in rechargeable batteries to be dispatched at a later period. There are several traditional uses cases for BESS, and it is typically used to enable intermittent renewable energy resources such as wind and solar by storing the energy to be deployed when there is greater demand, commonly referred to as load shifting. BESS can also be leveraged for arbitrage opportunities (charging when power prices are low and discharging when they are high), reducing curtailment of renewable resources, and ancillary services such as frequency balancing, which leverages BESS capabilities of ramping up and down within milliseconds to manage grid voltages. BESS is also leveraged for other capabilities such as peaking capacity and backup power. We note that the majority of demand for BESS is related to being paired with renewables, as ~50% of the solar in the interconnection queue today is paired with storage. However, with the rapid growth of data centers that is driving incremental power needs, BESS is being viewed as a unique solution given its many competitive attributes to other sources of power generation such as gas turbines or generators.

…and why do data centers want battery storage? Not only is the pace of data center deployments increasing, but power density is also increasing, driving incremental electricity demand. We highlight that data centers are not just getting larger, but the power compute needs are also rising exponentially. In particular, data centers have increased from 5-10 MW with power rack densities of 5-10 kw, to 100-200 MW today and in some cases GW-scale campuses or AI factories, with power rack densities of 50-100+ kw. As this load growth materializes, there remain questions on how to best meet this demand over the near-term as there remain certain bottlenecks regarding access to power such as interconnections, labor, and available supply. As a result, there is growing focus on behind-the-meter (BTM) generation solutions, particularly for gas and battery storage. In particular, our Utilities team forecasts that BTM power demand for data centers represents 0.5% of the 3.2% total power demand CAGR by 2030. This equates to ~3 GW of BTM DC power demand per year between 2027-2030. Importantly, BTM battery storage solutions are viewed as a near-term bridge, but also as a long-term backup option.

Exhibit 3: BTM data center opportunities represent 0.5% of the 3.2% power demand CAGR
Overall power demand CAGR by driver



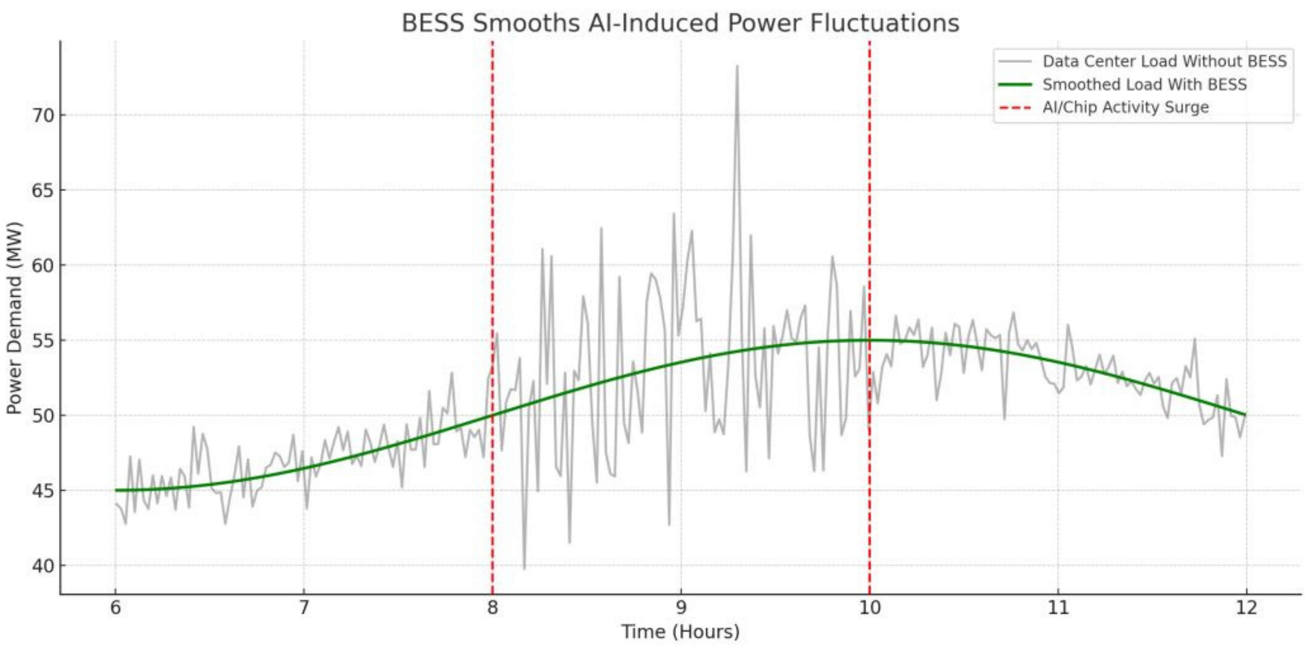
Source: Goldman Sachs Global Investment Research

Value proposition of BESS for data centers. Data centers, particularly those for hyperscalers focused on artificial intelligence leveraging language models with inference work loads, tend to have rapid power fluctuations based on usage trends. In particular, GPU's have the potential to go from 0% to 100% in milliseconds. While gas turbines are a superior solution for reliable baseload power, especially to match periods of peak demand, they are not ideal for more rapid changes in power demand. In generally, we hear from channel checks that gas turbines should ideally only shift power loads twice during a day, otherwise it risks wearing out engines at a more rapid pace. During a recent presentation from the Clean Energy States Alliance, we heard anecdotes of some new CCGT's breaking shafts in as little as seven months due to more frequent power load shifts. However, battery storage offers several advantages such as speed to market, power quality, backup power, and supporting on-site generation. These benefits are in addition to existing use cases around firming renewable power, load shifting, and voltage support on the grid. In particular, battery storage can be deployed within 12-18 months upon signing contracts, and is able to respond to power shifts within milliseconds, making it an ideal solution for AI DCs.

- **Speed to market.** Behind-the-meter (BTM) battery storage allows data center developers to bypass heavily congested utility interconnection queues, which can otherwise delay projects by four to eight years. By deploying on-site battery systems, operators can secure "first power" and begin phased tenant onboarding much earlier. This rapid deployment capability provides a competitive advantage for BESS, enabling hyperscalers to bring critical AI workloads online ahead of their competitors.
- **Rapid response.** Unlike traditional backup systems or reciprocating engines that require minutes to ramp up, BESS can deliver millisecond-level response times to manage sudden power disturbances. This instantaneous response allows BESS to

smooth out these rapid spikes from AI workloads, reducing strain on the grid. Furthermore, this rapid response capability also allows data centers to seamlessly participate in lucrative grid-balancing and frequency regulation markets.

Exhibit 4: BESS offers rapid response capabilities to support dynamic AI power usage
Overview of battery storage response to power fluctuations of AI data centers



Source: Fluence Energy, McKinsey

Gaining exposure to battery storage data center theme. Across our global research coverage, there are a number of companies with battery storage solutions that provide exposure to the data center end market. We include a non-exhaustive list of select stocks below with brief company descriptions on leverage to the opportunity.

- **CATL (Buy rated, covered by Nick Zheng)** dominates the global energy storage market with a ~30% market share at the battery level as of 2025 and explicitly identifies data centers as a core emerging application for its battery systems. The company provides high-capacity solutions like the TENER 6.25 MWh system and specialized products designed for data center energy management. CATL is also pioneering sodium-ion energy storage, with deliveries expected to begin in late 2026 to offer a cost-stable alternative to lithium-based systems. Its technology already supports major facilities, such as the SenseTime AI data center in Shanghai, where it has reduced electricity costs by 7%.
- **CSIQ (Sell rated, covered by Brian Lee)** operates in the data center storage market through its specialized subsidiary, e-STORAGE, which provides integrated BESS solutions like the SolBank 3.0 with 5 MWh to 8.36 MWh capacity units. The company is working with multiple data center customers to develop deeply integrated solar-plus-storage solutions, which it identifies as the most cost-effective way to power high-growth AI infrastructure. As of mid-2026, e-STORAGE manages a massive project development pipeline that includes 81 GWh of battery energy storage capacity across global markets. Recent contracts include a major deal with a

U.S. utility for a 2,493 MWh DC system specifically designed to support data center grid infrastructure and resiliency. We highlight that we expect battery storage to represent ~34% of total revenue for CSIQ in 2026, up from 3% in 2023 and 14% in 2024.

- **FLNC (Buy rated, covered by Brian Lee)** has secured an exclusive battery partnership in Nvidia's DSX Vera Rubin reference architecture, establishing it as a primary provider for AI computing infrastructure blueprints. The company has executed master supply agreements (MSAs) with two major hyperscalers and is actively bidding for orders for 12 GWh of capacity. Its Smartstack technology is specifically tailored for data centers, offering industry-leading density and advanced controls to manage the extreme power fluctuations typical of AI workloads. These solutions allow data center operators to bypass grid constraints, achieving faster "speed-to-power" while simultaneously providing backup power and voltage regulation. Importantly, as of F2Q26, FLNC maintains a 12 GW pipeline with data centers, which increased by 30% qoq, and we estimate that this opportunity could provide ~8%-14% to our F2028 revenue forecast.
- **F (Neutral rated, covered by Mark Delaney)** recently launched a wholly-owned subsidiary, Ford Energy LLC, to repurpose underutilized EV battery capacity for utility-scale storage systems targeting industrial and commercial customers, including data center demand. The company plans to invest $2bn to support a 20 GWh per year site in Kentucky, with potential for expansion, enabling it to provide a 5 MWh+ based energy storage systems, which the company expects to start shipping in late 2027. Ford will leverage its technology licensing partnership with CATL to manufacture LFP cells used in these store systems, which Ford expects will allow it to capture tax credits associated with domestic production. The autos team estimates that Ford Energy will contribute ~$3 bn in revenue in 2028, relative to ~$205 bn for the total company (inclusive of Ford Energy), though this could account for a larger portion of EBIT depending on the profitability (as Energy storage at scale typically carries better margins vs Ford's traditional auto business).
- **LGES (Buy rated; Covered by Nikhil Bhandari)** is positioning itself as a "one-stop shop" for data centers by integrating battery hardware with software-based system integration (SI) through its subsidiary, Vertech. The company recently debuted a next-generation lithium iron phosphate (LFP) UPS battery that delivers over double the power output of existing products, significantly improving the density and space requirements for high-performance computing environments. To meet surging demand, LG is converting five of its North American production bases into dedicated ESS lines and expects to reach 50 GWh of North American capacity by the end of 2026. Its "One LG" strategy further enhances this exposure by bundling battery systems with the group's proprietary liquid cooling and data center design services. Storage is becoming a structurally larger part of the mix, with ESS revenue rising to c.31%/c.35% by 2026E/2027E (from c.14% in 2025E), driven by BTM storage opportunities.
- **NXT (Buy rated (on Americas CL), covered by Brian Lee)** is transforming from a solar tracker supplier into a comprehensive energy technology platform through its announced acquisition of Prevalon Energy and its 6+ GWh of deployed and

contracted BESS capacity. The company's new storage platform features a Hybrid Power Stabilizer designed to help manage the fast, variable load swings and GPU workload smoothing required by AI data centers. By integrating these storage capabilities with its internally developed power conversion systems, NXT aims to deliver a complete "everything but the panel" solution for hyperscale developers. Importantly, management has discussed that Prevalon is already executing on an AI DC project with a tier one hyperscaler. While the acquisition of Prevalon has not yet closed, management has outlined over $300mn in annualized revenue, which could represent ~7% of consolidated revenue in F2028, and not including any cross-selling opportunities with its other product categories.

- **Samsung SDI (006400.KS, Neutral rated, covered by Nikhil Bhandari)** targets the AI DC market with specialized high-power batteries for UPS and Battery Backup Units (BBU) that are becoming standard in server racks for major cloud providers like GOOGL, META, and AMZN. The company's U8A1 battery uses a proprietary prismatic form factor and LMO chemistry to achieve a 33.0% improvement in space efficiency for data center facilities. Additionally, the Samsung Battery Box (SBB) 1.5 provides a containerized solution using high-nickel NCA cells for large-scale energy management installations. ESS is an increasingly meaningful driver, with ESS battery revenue rising to c.26%/c.28% of total sales in 2026E/27E (from c.21% in 2025), driven by BTM storage opportunities.
- **SHLS (Buy rated, covered by Brian Lee)** is exposed to the data center storage market as the leading provider of electrical balance of systems (EBOS), which serves as the "central nervous system" connecting battery blocks to the power grid. The company has adapted its high-voltage DC power distribution expertise to create a BESS Recombiner platform specifically for data centers, allowing for the aggregation of multiple power sources into a single, stable output. By using prefabricated, "plug-and-play" components, Shoals helps data center operators complete projects faster and with less skilled on-site labor, addressing critical bottlenecks. The company is actively developing additional tailored offerings for data center power systems, viewing the sector as a high-margin opportunity to diversify its core solar infrastructure business. Management estimates that its BESS products maintained $67mn in backlog and awarded orders (BLAO) as of 4Q25 or 9% of its ~$750mn in total BLAO and anticipates its BESS products can reach ~$30mn-$35mn in revenue in 2026, or ~5% of total revenue.
- **Sungrow (Neutral rated, covered by Jacqueline Du)** leverages its position as a leading global PV inverter and energy storage provider to deliver integrated "grid-to-chip" solutions specifically designed for the next generation of AI data centers. The company recently released an AI Data Center Full-chain Power Supply Solution White Paper outlining its "Green-Gray-White" three-space architecture designed to support gigawatt-scale clusters and megawatt-level racks. Its flagship PowerTitan 3.0 and industrial-grade PowerStack series provide the high-density liquid-cooled storage necessary to manage the rapid load swings and 24/7 reliability required by hyperscalers. Additionally, the company is launching dedicated Solid-State Transformer (SST) solutions to reduce the physical footprint of power infrastructure within data center facilities while improving overall efficiency. As of

2025, Sungrow generated 42%/49% sales/gross profits from energy storage systems.

- **TSLA (Neutral rated, covered by Mark Delaney)** maintains an energy storage segment that is a key beneficiary of rapid load growth from AI, with its Megapack and Megablock products serving as critical grid-stabilizing infrastructure for hyperscalers. To support this growth, TSLA deployed a record 46.7 GWh of storage in 2025 (and expects 2026 deployments to exceed this level) and is ramping up production at a new Megafactory near Houston. Separately, the company recently announced a distributed power pact with RUN and Renew homes to aggregate home batteries and smart thermostats to provide 16.8 GW of flexible capacity specifically for utilities and data center operators. This framework allows data centers to draw on residential storage during peak demand, avoiding the 3-6 year lead times typically required for grid capacity expansions. In 2028, the autos team estimates that Tesla's Energy business (which encompasses both its commercial Megapack/Megablock products and the retail Powerwall product) will account for $29 bn of Tesla's $140 bn in revenue and $8.5 bn of the total $29.6 bn gross profit.

Long duration energy storage (LDES) and other technologies. Lithium ion remains the most commercially advanced BESS technology, which has helped to dramatically bring down capital costs over the past several years. However, there are other storage technologies looking to come to market to address the power demand needs of data centers, albiet not all in BTM settings. This includes redox flow batteries as well as other long duration storage solutions (LDES) such as compressed air, sodium-ion batteries, and hydrogen fuel cells.

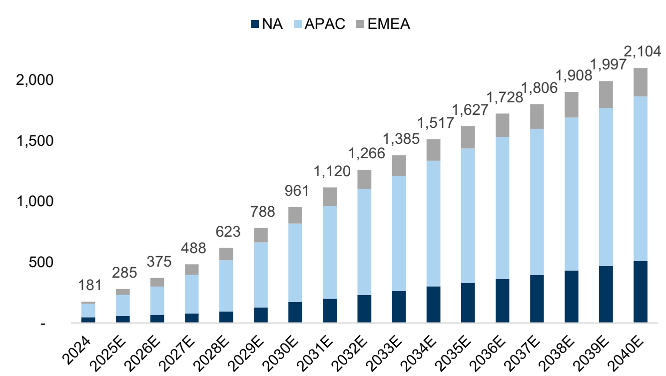
- **Redox flow and LDES.** Redox flow batteries, such as vanadium, store energy in liquid electrolytes housed in tanks. These water-based chemistry solutions maintain no thermal runaway risks, and provide 8-12+ hours of BTM backup power, which helps facilities with grid outages while managing peak-shaving. Key companies pioneering this space for data centers include Sumitomo Electric, Storion Energy (a JV of Stryten Energy and Largo Clean Energy), XL Batteries, and FlexBase. Other long duration energy storage (LDES) include compressed air solutions from companies like Hydrostor, pumped hydro storage, and metal-air batteries such as iron-air batteries from Form Energy.
- **Sodium-ion batteries.** Sodium-ion batteries utilize abundant sodium instead of lithium, offering a highly sustainable and cost-effective battery storage solution. Prominent companies developing and deploying this technology for data centers include Peak Energy, Inlyte Energy, and ABB. These solutions deliver high peak-power discharges and operating across a wide temperature range, which significantly reduces the space and cooling requirements compared to traditional systems. Additionally, they are intrinsically safe with zero risk of thermal runaway or combustion, allowing them to be safely deployed within IT equipment racks to manage volatile AI workloads.
- **Hydrogen fuel cells.** Hydrogen fuel cells generate clean electricity through a chemical reaction between hydrogen and oxygen. Major players driving hydrogen fuel cell integration in data centers include BE, PLUG, BLDP, CAT, and VRT. These

systems offer modular designs to scale power output, and address data center power demand by serving as a highly reliable, zero-emission alternative to traditional diesel backup generators.

Updating battery storage supply/demand model. We update estimates for our battery storage supply/demand to reflect new forecasts for BTM storage applications for DCs. Additionally, we have made some slight changes to our US demand forecasts, while increasing forecasts in Australia and Europe based on current market dynamics.

- **US demand.** We now include BTM storage deployments, with forecasts of ~60 GWh in 2030. We also make some minor adjustments to our forecasts for the resi end market. As a result, we now anticipate 172 GWh in 2030, up from 112 GWh.
- **Europe/EMEA.** We revise forecasts in Europe as well as the Middle East and Africa. In particular, we are now forecasting battery storage deployments in EMEA of 142 GWh in 2030, up from 83 GWh. The regions experiencing the largest growth are Germany, Spain/Portugal, and other regions in EMEA.
- **Australia.** We are raising storage deployments in Australia mostly as a result of positive policy measures within the residential end market, with our forecasts more than doubling in 2026 through 2028, as policy support is expected to gradually decline through 2030.

Exhibit 5: We forecast global energy storage installs to reach ~2,100GWh in 2040
Annual battery energy storage intsalls by region, GWh, 2024-2040E



Source: Goldman Sachs Global Investment Research

Exhibit 6: We forecast 30% yoy growth in global battery energy storage installations in 2027E
Annual battery energy storage installs by region and country, MWh 2024-2040E

Batteries Demand (MWh)	2024	2025E	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E	2038E	2039E	2040E
Asia Pacific																	
Japan	2,239	2,528	2,863	3,381	3,971	4,396	4,877	5,178	5,498	5,838	6,199	6,583	6,992	7,426	7,888	8,380	8,902
S. Korea	955	1,109	1,210	1,415	1,498	1,685	1,787	2,052	2,290	2,554	2,846	3,169	3,576	3,969	4,403	4,881	5,407
China	101,575	155,352	211,003	286,917	381,264	492,703	590,179	693,233	780,752	851,929	931,656	997,717	1,053,628	1,081,617	1,129,219	1,162,801	1,210,436
India	2,149	3,674	6,763	12,904	20,244	30,097	41,498	56,034	73,565	77,760	82,193	86,878	91,829	97,061	102,590	108,434	114,609
Australia	3,755	10,669	11,459	12,369	12,933	7,010	8,135	8,876	9,687	10,574	11,544	12,606	13,284	14,001	14,757	15,554	16,396
APAC total	**110,674**	**173,332**	**233,297**	**316,986**	**419,909**	**535,890**	**646,476**	**765,374**	**871,792**	**948,654**	**1,034,438**	**1,106,953**	**1,169,309**	**1,204,074**	**1,258,857**	**1,300,049**	**1,355,750**
Growth	94%	57%	35%	36%	32%	28%	21%	18%	14%	9%	9%	7%	6%	3%	5%	3%	4%
Europe, Middle East, Africa																	
Germany	5,962	9,356	12,437	13,994	15,387	17,297	18,926	20,233	21,458	22,760	24,144	25,616	27,182	28,847	30,618	32,502	34,508
Italy	3,713	5,454	6,503	7,485	8,411	11,102	12,285	13,001	13,597	14,228	14,898	15,608	16,363	17,164	18,015	18,920	19,882
Spain & Portugal	497	719	1,957	3,263	4,393	5,701	6,546	7,369	8,220	9,100	10,012	10,958	11,942	12,966	14,034	15,149	16,316
UK	5,631	5,937	8,600	9,835	11,302	12,631	14,225	15,875	17,729	19,814	22,160	24,802	27,781	31,143	34,939	39,231	44,086
EMEA - other	8,149	34,130	45,382	57,668	67,931	78,299	90,353	99,535	102,952	106,374	110,042	113,220	115,445	117,725	120,064	122,463	124,924
EMEA total	**23,951**	**55,595**	**74,879**	**92,245**	**107,424**	**125,029**	**142,335**	**156,013**	**163,955**	**172,276**	**181,255**	**190,205**	**198,712**	**207,844**	**217,670**	**228,265**	**239,716**
Growth	40%	132%	35%	23%	16%	16%	14%	10%	5%	5%	5%	5%	4%	5%	5%	5%	5%
North America																	
USA	46,621	56,472	66,721	78,908	95,588	127,353	172,180	199,061	230,184	263,822	301,512	329,715	360,422	394,231	431,520	468,533	508,993
Residential	2,899	4,462	3,875	4,458	4,934	5,271	5,703	6,345	6,734	7,173	7,748	8,365	8,772	9,199	9,649	9,892	10,142
Non-residential	1,329	1,614	2,405	2,969	3,681	4,420	5,031	5,459	5,781	6,123	6,485	6,869	7,271	7,648	8,044	8,379	8,727
Utility	42,393	50,395	60,441	71,481	86,973	117,662	161,445	187,257	217,669	250,526	287,279	314,481	344,379	377,384	413,828	450,262	490,123
Canada																	
North America total	**46,621**	**56,472**	**66,721**	**78,908**	**95,588**	**127,353**	**172,180**	**199,061**	**230,184**	**263,822**	**301,512**	**329,715**	**360,422**	**394,231**	**431,520**	**468,533**	**508,993**
Growth	64%	21%	18%	18%	21%	33%	35%	16%	16%	15%	14%	9%	9%	9%	9%	9%	9%
Total	**181,246**	**285,400**	**374,898**	**488,139**	**622,922**	**788,272**	**960,991**	**1,120,448**	**1,265,931**	**1,384,752**	**1,517,206**	**1,626,874**	**1,728,443**	**1,806,150**	**1,908,047**	**1,996,847**	**2,104,459**
Growth	77%	57%	31%	30%	28%	27%	22%	17%	13%	9%	10%	7%	6%	4%	6%	5%	5%
Global cell/Batteries capacity (MWh)	459,168	648,374	902,978	1,060,667													
Utilization rate	46%	52%	48%	50%													

Source: Goldman Sachs Global Investment Research

Stock implications from battery storage opportunities

We see Buy rated **FLNC** as a meaningful beneficiary of the growing focus on BTM storage solutions, especially for data centers. In particular, the company has a 12 GWh pipeline directly tied to data centers and is actively bidding on two master supply agreements with two hyperscalers that represents a majority of this pipeline and is likely to start being realized in shipments by the end of F2027. As a result, we see upside potential to FLNC's F2028 estimates. We estimate that if FLNC can secure 15%-25% of the 12 GWh pipeline, this could represent ~1.8 GWh - 3.0 GWh of storage opportunity. Assuming a ~$200/kWh price, this represents $360mn-$600mn in revenue opportunity, or ~8%-14% upside to our F2028 revenue forecast. Additionally, we believe this 12 GWh pipeline is likely to increase over time, providing further upside to this estimate.

We also see Sell rated **NRGV** as able to capture incremental demand for Asset Vault as well as its core battery integration business. Within Asset Vault's powered land and powered shell segments, the company uniquely serves hyperscalers and data center companies. The company has already laid out targets of reaching ~350 MW and ~1.8 GW of powered shell and powered land projects under construction and development by 2030. Separately, the company's core battery integration business could also see upside from incremental demand for battery storage. However, we continue to look for the Asset Vault to expand its operating portfolio and gain more traction before we get more positive on the outlook for the business.

Updating estimates for NRGV

We are updating estimates and valuation methodology for NRGV given its continued progress advancing its Asset Vault business. In particular, we have revised our outlook for Asset Vault, including adding in new projects to our forecasts, while making additional changes to our revenue assumptions. At the same time, we are taking a more conservative view on the outlook for NRGV's software and services segment given recent topline trends. As a result, our margin forecasts for 2026 and 2027 have declined by ~400bps, while our margins in 2028 expanded by ~400 bps. We note that our gross margin forecast of 15.9% in 2026 remains at the low-end of managements 15%-25% guidance range.

Incorporating valuation for Asset Vault. Given NRGV's continued efforts to establish its Asset Vault business, we are now incorporating incremental value for the segment in our consolidated company valuation. We value the company at a 6.0X EV/EBITDA multiple on our forecasted 2028 EBITDA generation for the segment. We view this multiple as ~3.5x below our comp set average, which we attribute to the still relatively early stage nature of the business. Additionally, we estimate a ~64% EBITDA margin for Asset Vault in 2028, which meaningfully exceeds the comp group average. In aggregate, this equates to a $1.75/sh valuation.

Exhibit 7: We currently value Asset Vault below see IPP comps as providing re-rating potential as EBITDA expands beyond 2030
IPP comps vs NRGV

			EV/EBITDA				EBITDA Margin				
Company	Ticker	EV ($mn)	2027	2028	2029	2030	2027	2028	2029	2030	2027-30 Rev CAGR
Talen Energy Corp	TLN	23,699	8.9X	8.1X	7.7X	7.3X	52%	54%	53%	54%	6%
Vistra Corp.	VST	75,478	9.0X	8.4X	7.7X	7.4X	33%	34%	38%	39%	1%
Constellation Energy Corporation	CEG	108,670	11.5X	9.8X	9.2X	8.9X	27%	30%	30%	31%	5%
NRG Energy, Inc.	NRG	53,102	8.7X	8.2X	7.9X	7.4X	17%	17%	18%	19%	3%
NextEra Energy, Inc.	NEE	296,297	13.8X	12.6X	12.6X	11.1X	63%	64%	60%	63%	8%
Average			**10.4X**	**9.4X**	**9.0X**	**7.9X**	**37%**	**39%**	**40%**	**41%**	**4%**
Median			**9.0X**	**8.4X**	**7.9X**	**7.4X**	**33%**	**34%**	**38%**	**39%**	**5%**
Energy Vault Holdings, Inc.	NRGV	836	92.7X	11.9X	6.0X	4.3X	3%	17%	25%	25%	36%

Priced as of 7/14/26

Source: FactSet, Goldman Sachs Global Investment Research

Estimate changes. Our 2026E-2028E revenue moves by (3%)/(1%)/5% to $243mn/$303mn/$364mn, with our adj. EBITDA moving to ($26mn)/($10mn)/$32mn from ($14mn)/$9mn/$33mn, mostly due to changes to our forecast for its core software and services segments as well as changes to our Asset Vault assumptions.

Valuation. Our 12-month price target of $2.50 (from $2.00) is now based on an 0.8X EV/S multiple (from 1.4X due to lower comps and a more cautious outlook on margins) on 2027E revenue of its core battery integration business, yielding a $1.25 valuation, and a 6.0X EV/EBITDA multiple on our Asset Vault EBITDA in 2028, yielding a $1.75 valuation. We then subtract net debt of $0.50/sh to arrive at our $2.50 price target. We are no longer using a DCF analysis as part of our valuation as we believe our new methodology is more reflective of the overall business.

Key risks. Key risks include higher than expected market demand, decrease in customer concentration, and faster than expected cost declines.

Disclosure Appendix

Reg AC

We, Brian Lee, CFA, Tyler Bisset, CFA and Keshav Choudhary, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

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Contributing Authors: Brian Lee, CFA Goldman Sachs & Co. LLC, Tyler Bisset, CFA Goldman Sachs & Co. LLC, Keshav Choudhary Goldman Sachs India SPL.

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The rating(s) for Energy Vault Holdings is/are relative to the other companies in its/their coverage universe: Acuity Inc., Array Technologies Inc., Cameco Corp., Canadian Solar Inc., Energy Vault Holdings, Enphase Energy Inc., First Solar Inc., Fluence Energy Inc., HA Sustainable Infrastructure, Hayward Holdings, JinkoSolar Holding, MP Materials, Mueller Water Products Inc., Nextpower, Nuscale Power Corp., Oklo Inc., Pentair Plc, Ramaco Resources Inc., Shoals Technologies, SolarEdge Technologies Inc., Sunrun Inc., Universal Display Corp., Uranium Energy Corp., Veralto Corp., Watts Water Technologies Inc., Wolfspeed Inc., Xylem Inc.

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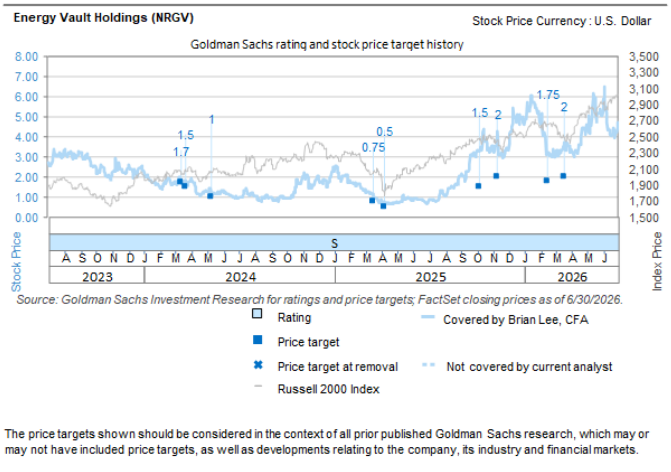
Price target and rating history chart(s)



Target price history table(s)

Energy Vault Holdings (NRGV)

Date of report	Target price ($)	Closing price ($)
16-Jul-26	2.50	3.04
18-Mar-26	2.00	3.59
16-Feb-26	1.75	3.19
11-Nov-25	2.00	4.29
07-Oct-25	1.50	3.31
08-Apr-25	0.50	0.67
18-Mar-25	0.75	0.90
10-May-24	1.00	1.12
22-Mar-24	1.50	1.66
13-Mar-24	1.70	1.90

Price targets shown in table(s) are unadjusted for corporate actions.

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